

COURSE TITLE : TECHNOLOGY AND MEDICAL SCIENCE
COURSE CODE : 5244
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER : 52
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Introduction to Biotechnology	13
2	Introduction to Bioinformatics	13
3	Hospital Information System	13
4	Internet in Medicine	13
	TOTAL	52

Course General Outcome :

SL.NO	SUB	STUDENT WILL BE ABLE TO
1	1	To understand Biotechnology and related terms
2	2	To understand Bioinformatics
	3	To comprehend Artificial Intelligence and its applications
3	4	To understand hospital automation
	5	To comprehend hospital information system
4	6	To Apply Internet in Medicine

G.O. : General Outcome

On completion of the study the student will be able:

MODULE I INTRODUCTION TO BIOTECHNOLOGY

1.1.0 To understand Biotechnology and related terms

- 1.1.1 To define Biotechnology and Genetic Engineering
- 1.1.2 To explain the basic concepts of molecular biology, cells and chromosomes.
- 1.1.3 To explain the structure of DNA, RNA and proteins.
- 1.1.4 To describe DNA as the genetic material & Role of genes within cells
- 1.1.5 To explain mechanism of DNA replication & Characteristics of gene mutation
- 1.1.6 To explain the Basic tools of genetic engineering
- 1.1.7 To explain fundamentals of synthesis of various industrial products by genetic Engineering
- 1.1.8 To describe cloning and transgenic animals.

MODULE II INTRODUCTION TO BIOINFORMATICS

2.1.0 To understand Bioinformatics

- 2.1.1 To define Bioinformatics and computational biology
- 2.1.2 To state the importance of Bioinformatics

- 2.1.3 To define Biological database
- 2.1.4 To list the different types of Biological database
- 2.1.5 To explain the basics of Protein Data bank
- 2.1.6 To illustrate Computer Aided Drug Design and Genome mapping

2.2.0 To comprehend Artificial Intelligence and its applications

- 2.2.1 To define Artificial Intelligence
- 2.2.2 To explain the disciplines and techniques used in AI
- 2.2.3 To explain the creation of knowledge base
- 2.2.4 To explain logical reasoning and decision making.
- 2.2.5 To give application examples of expert system using AI- MYCIN, EMYCIN.

MODULE III HOSPITAL INFORMATION SYSTEM

3.1.0 To understand hospital automation

- 3.1.1 To study the concept of hospital automation.
- 3.1.2 To explain the concept of automated clinical laboratories
- 3.1.3 To describe the automated methods in hematology
- 3.1.4 To explain the chromosome analysis by computer
- 3.1.5 To explain computerised ECG, EMG.

3.2.0 To comprehend hospital information system

- 3.2.1 To state the concept of Hospital Information System
- 3.2.2 To list the functional capabilities of computerized hospital information system
- 3.2.3 To state the concept of web based Information system
- 3.2.4 To list the benefits of computerised patient record
- 3.2.5 To explain Portable Personal health record
- 3.2.6 To describe computerised prescription

MODULE IV INTERNET IN MEDICINE

4.1.0 To Apply Internet in Medicine

- 4.1.1 To describe the use of online services in medicine
- 4.1.2 To list the security issues
- 4.1.3 To describe confidentiality through cryptography.
- 4.1.4 To describe the concept of digital signature
- 4.1.5 To describe the concept of user authentication
- 4.1.6 To describe the basic principles of telemedicine and Video conferencing
- 4.1.7 To list the applications in telemedicine
- 4.1.8 To explain tele-surgery
- 4.1.9 To explain ICU ambulance
- 4.1.10 To explain medical data sharing through internet.

CONTENTS

MODULE I BIOTECHNOLOGY AND RELATED TERMS

Definition of Biotechnology and Genetic Engineering - Concepts of molecular biology, cells and chromosomes- Structure of DNA, RNA and proteins-DNA as the genetic material - Role of genes within cells- Mechanism of DNA replication - Characteristics of gene mutation- Tools of genetic engineering- Fundamentals of synthesis of various industrial products by genetic Engineering-Cloning and transgenic animals.

MODULE II BIOINFORMATICS

Definition of Bioinformatics and computational biology-Importance of Bioinformatics
Definition and types of Biological database- Protein Data bank-Computer Aided Drug Design and Genome mapping
Artificial Intelligence and its applications
Definition of Artificial Intelligence - disciplines and techniques used in AI- creation of knowledge base- logical reasoning and decision making- MYCIN- EMYCIN.

MODULE III HOSPITAL AUTOMATION

Concept of hospital automation and automated clinical laboratories - Automated methods in haematology - Chromosome analysis by computer - Computerised ECG, EMG
Hospital information system
Concept of Hospital Information System- Functional capabilities of computerized hospital information system - Concept of web based Information system -Benefits of computerised patient record - Portable Personal health record- Computerised prescription

MODULE IV INTERNET IN MEDICINE

Use of online services in medicine-Security issues- Confidentiality through cryptography - concept of digital signature, user authentication-Principles of telemedicine and Video conferencing - Applications in telemedicine -Tele-surgery- ICU ambulance- Medical data sharing through internet.

TEXT BOOK

- | | |
|----------------------------|--|
| 1.Essential Bioinformatics | -Jin Xiong- Cambridge University Press-A1Books |
| 2. Computers in Medicine | -R D Lele- Tata McGraw-Hill Pub., 1988 |

REFERENCES

1. Artificial Intelligence –A modern approach –Stuart Russel, Peter Norvig- Pearson, 2014
- 2.Handbook of Biomedical Instrumentation- RS Khandpur- Tata McGraw-Hill Pub., 2006